QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/401A
9 August 1983
SUPERSEDING
MIL-M-38510/401(USAF)
22 October 1981

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON-GATE, MONOLITHIC PERIPHERAL INTERFACE ADAPTER

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION

This specification is approved for use by Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic N-channel, silicon gate peripheral interface adapter. One product assurance class and a choice of case outlines and lead finishes are provided for each type, and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510 with the exception that the "JAN" or "J" certification mark shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Microprocessor peripheral interface adapter

1.2.2 Case outline. The case outline shall be designed as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
Q	D-5 (40-lead, 9/16" x 2-1/16"), dual-in-line package

1.3 Absolute maximum ratings.

V_{CC} supply voltage range	-0.3 to +7.0 V dc
Input voltage range	-0.3 to +7.0 V dc
Storage temperature range	-55°C to +150°C
Power dissipation (P_D)	550 mW
Lead temperature (soldering 5 seconds)	270°C
Maximum junction temperature (T_J)	150°C
Thermal resistance, junction to case (θ_{JC})	32.5°C/W

1.4 Recommended operating conditions.

Supply voltage	4.75 V dc to 5.25 V dc
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.8 V dc
Frequency of operation	0.01 to 1 MHz
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

SPECIFICATION

MILTARY

MIL-M-38510 - Microcircuits, General Specification For.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510 and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional diagram. The functional diagram shall be as specified on figure 2.

3.2.3 Functional description, terms, and symbols. The functional description, terms, and symbols shall be as specified in 6.5.

3.2.4 Case outline. The case outline shall be as specified in 1.2.2.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510. (See 6.6 of this spescification.)

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturers eligibility. To be eligible to supply microcircuits to this specification a manufacturer shall have a manufacturer's certification in accordance with MIL-M-38510 for at least one line. Not necessarity the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

V_{CC} = 5.0 V ±5%

T_C = -55 to +125°C (unless otherwise specified)

Test	Symbol	Conditions	Figure	Limits Min	Limits Max	Unit
Quiescent supply current	I_{CC}	V_{CC} = 5.25 V dc V_{IN} = 0, 5.25 V dc	5		100	mA
Input leakage current R/W, $\overline{RESET}$, RS0, RS1, CS0, CS1, $\overline{CS2}$, CA1, CB1, ENABLE	I_{IH1}	V_{CC} = 5.25 V dc V_{IN} = 5.25 V dc			2.5	µA
Input high current PA0-PA7, CA2	I_{IH2}	V_{CC} = 5.25 V dc V_{IH} = 2.4 V dc		-200		µA
Input low current PA0-PA7, CA2	I_{IL1}	V_{CC} = 4.75 V dc V_{IL} = 0.4 V dc			-2.4	mA
Output leakage current high (off state) $\overline{IRQB}$, $\overline{IRQA}$	I_{LOH}	V_{CC} = 4.75 V dc V_{OH} = 2.4 V dc			10	µA
Output leakage current low (off state) $\overline{IRQB}$, $\overline{IRQA}$	I_{LOL}	V_{CC} = 4.75 V dc V_{OL} = 0.4 V dc			-10	µA
Output high voltage D0-D7	V_{OH1}	V_{CC} = 4.75 V dc I_{OH} = -205 µA		2.4		V
Output high voltage PA0-PA7, CA2 PB0-PB7, CB2	V_{OH2}	V_{CC} = 4.75 V dc I_{OH} = -200 µA		2.4		V
Output high voltage PA0-PA7, CA2	V_{OH3}	V_{CC} = 4.75 V dc I_{OH} = -10 µA		3.75		V
Output low voltage D0-D7	V_{OL1}	V_{CC} = 4.75 V dc I_{OL} = 1.6 mA			0.4	V
Output low voltage PA0-PA7, CA2 PB0-PB7, CB2 $\overline{IRQB}$, $\overline{IRQA}$	V_{OL2}	V_{CC} = 4.75 V dc I_{OL} = 1.6 mA			0.4	V
Darlington drive current PB0-PB7, CB2	I_{OH1}	V_{CC} = 4.75 V dc V_O = 1.5 V dc		-1.0	-10	mA
Three-state (off state) input current PB0-PB7, CB2, D0-D7	I_{IL2}	V_{CC} = 4.75 V dc V_{IN} = 0.4 V dc			10.0	µA
	I_{IH3}	V_{CC} = 4.75 V dc V_{IN} = 2.4 V dc		000	10.0	µA
Input capacitance R/$\overline{W}$, CS0, $\overline{CS2}$, CS1, ENABLE, $\overline{RESET}$, RS1, RS0	C_I	V_{CC} = V_{SS} = GND V_{IN} = 0, f = 1.0 MHz			7.5	pF
D0-D7					12.5	pF
PA0-PA7, CA2, CA1, PB0-PB7, CB1, CB2					10.0	pF
Output capacitance $\overline{IRQB}$, $\overline{IRQA}$	C_{OUT}				5.0	pF

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.
V_{CC} = 5.0 V ±5%
T_C = -55 to +125°C (unless otherwise specified)

Test	Symbol	Conditions	Figure	Limits Min	Limits Max	Unit
Enable cycle time	t_{cycE}	V_{CC} = 5.0 V dc, V_{SS} = GND, C = 130 pF, R = 11.7 kΩ, R_L = 2.5 kΩ	7,10	1.0		µs
Enable pulse width, high	t_{PWH1}			450		ns
Enable pulse width, low	t_{PWL1}			430		ns
Input clamp voltage	V_{IC}	(POS) V_{IN} = +5.25 V dc			2.5	µA
		(NEG) I_O = 10 µA			-0.9	V
Setup time CS0, CS1, CS2 RS0, RS1 and R1W valid to enable positive transition	t_{SLH1} t_{SHH1}	C = 130 pF, R = 11.7 kΩ R_L = 2.5 kΩ	7,11,12	160		ns
Address hold time	t_{HLH1} t_{HLL1}			10		ns
Data delay time, READ	t_{PZL1} t_{PZH1}		7,11		320	ns
Data hold time, READ	t_{HHZ1} t_{HLZ1}			10		ns
Data setup time, WRITE	t_{SZL1} t_{SZH1}		7,12	195		ns
Data hold time, WRITE	t_{HLH3} t_{HLL3}			10		ns
Peripheral data setup time	t_{SHH2} t_{SLH2}	C = 40 pF, R = 12 kΩ (Adjust R_L so that I_I = 3.2 mA with	7,13	200		ns
Peripheral data hold time	t_{HLH4} t_{HLL4}	V_I = 0.4 V dc and V_{CC} = 5.0 V dc.)		0		ns
Delay time enable negative transition to CA2 negative transition	t_{PLL1}		7,14,15		1.0	µs
Delay time enable negative transition to CA2 positive transition	t_{PLH1}		7,14			µs
Delay time from CA1 active transition to CA2 positive transition	t_{PLH2}		7,15		2.0	µs
Delay time enable negative transition to peripheral data valid	t_{PLH3} t_{PLL3}	C = 40 pF, R = 12 kΩ (Adjust R_L so that I_I = 3.2 mA with V_I = 0.4 V dc)	7,16,17		1.0	µs
Delay time enable negative transition to peripheral CMOS data valid PA0-PA7, CA2	t_{PLH4} t_{PLL4}	C = 30 pF	8,16		2.0	µs

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.
V_{CC} = 5.0 V ±5%
T_C = -55 to +125°C (unless otherwise specified)

Test	Symbol	Conditions	Figure	Limits Min	Limits Max	Unit
Delay time enable positive transition to CB2 negative transition	t_{PHL5}	C = 40 pF, R = 12 kΩ (Adjust R_L so that I_I = 3.2 mA with V_I = 0.4 V dc)	7,18,19		1.0	µs
Delay time peripheral data valid to CB2 negative transition	t_{PHL6}		7,17	20		ns
Delay time enable positive transition to CB2 positive transition	t_{PHH5}		7,18		1.0	µs
Delay time, CB1 active transition to CB2 positive transition	t_{PLH6}		7,19		2.0	µs
Peripheral control output pulse width, CA2/CB2	t_{PWL2}		9,14,18	550		ns
Delay time from enable to $\overline{IRQA}$, $\overline{IRQB}$ release	t_{PHH7}	C = 100 pF R = 3 kΩ	9,21		1.60	µs
Interrupt response time	t_{PHL7}		9,20		1.0	µs
Interrupt input pulse width	t_{PWL3} t_{PWH3}		20	500		ns
Reset low time	t_{PWL4}		22	1.0		µs

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 6, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in, divided by the total number of devices submitted for burn-in in that lot, shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be as those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.7) may be used to satisfy the requirement for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroup 4 (C_j, measurement) shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 . Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 6 or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroup (see table III) Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1,7
Final electrical test parameters (method 5004)	*1,2,*7,8
Group A test requirements (method 5005) 1/	1,2,4,7,8,12
Group B V_{ZAP} test	See 4.5.3
Group C end-point electrical parameters (method 5005)	1,2,7
Additional electrical subgroups for group C periodic inspections	None

*PDA applies to subgroups 1 and 7 (see 4.2c).

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. When the devices are measured at 25°C following application of the steady-state or burn-in test condition, they shall be cooled to 35°C prior to removal of the bias.

4.5.3 High voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse of 400 volts from a 100 pF source in the following test sequence:

a. Measure I_{IH} and I_{IL} at one input terminal of the device under test at 25°C. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be as specified in table III.

b. Apply the test voltage (V_{ZAP} = 400 V) to the same terminal selected for input current measurements. Apply V_{ZAP} in a 2 pulse sequence as follows and in accordance with the test circuit of figure 4.

(1) Input (-) to V_{SS}.

(2) Input (+) to V_{SS}.

c. Within 24 hours repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. At this time a DUT exhibiting input currents in excess of the specified limits is defective.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be as specified in MIL-M-38510. The devices covered by this specification require electrostatic protection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

Device type 01

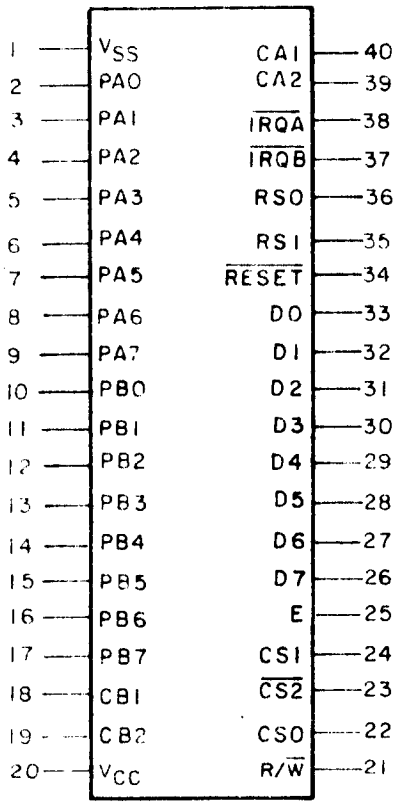


FIGURE 1. Terminal connections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

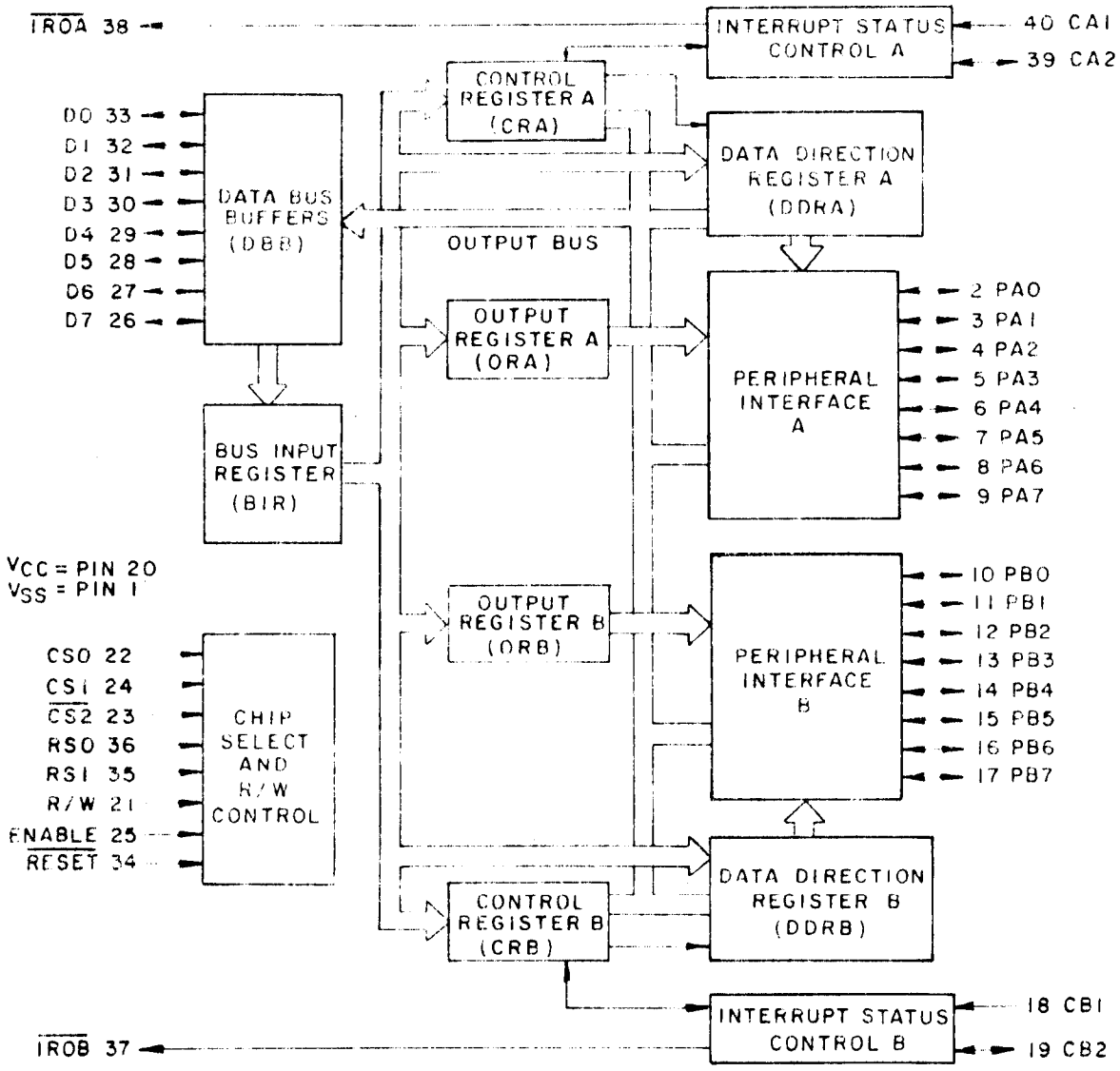
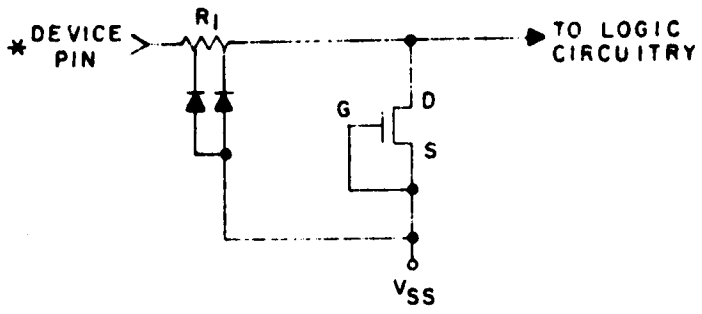


FIGURE 2. Functional diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

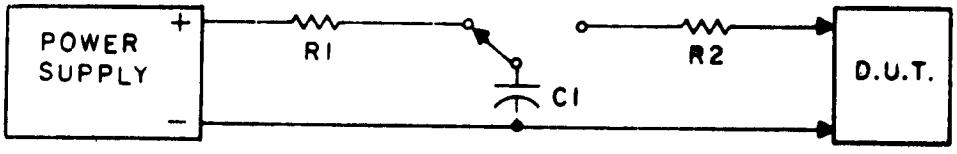


R_1 = 1k to 2k typical for all inputs.

$B_{V_{DSS}}$ = 20V to 30V typical.

*Each input has an input protection network

FIGURE 3. Input protection network.

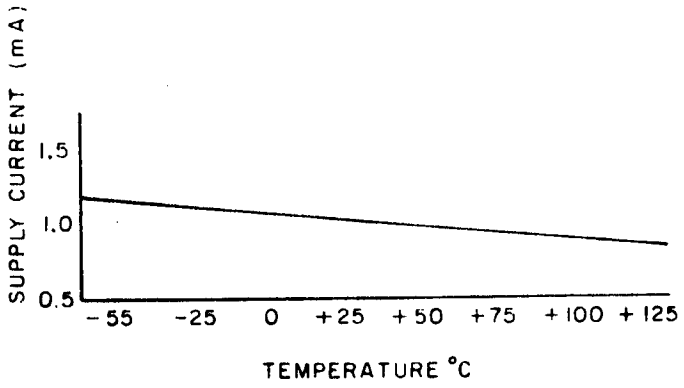


R1 = 10 MΩ to 50 MΩ
R2 = 1.5 kΩ ±10%
C1 = 100 pF ±20%
V_{ZAP} = 400 V measured at device input terminal

Mode	+ Terminal	- Terminal
1	V_{SS}	Input
2	Input	V_{SS}

FIGURE 4. High voltage (V_{ZAP}) test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

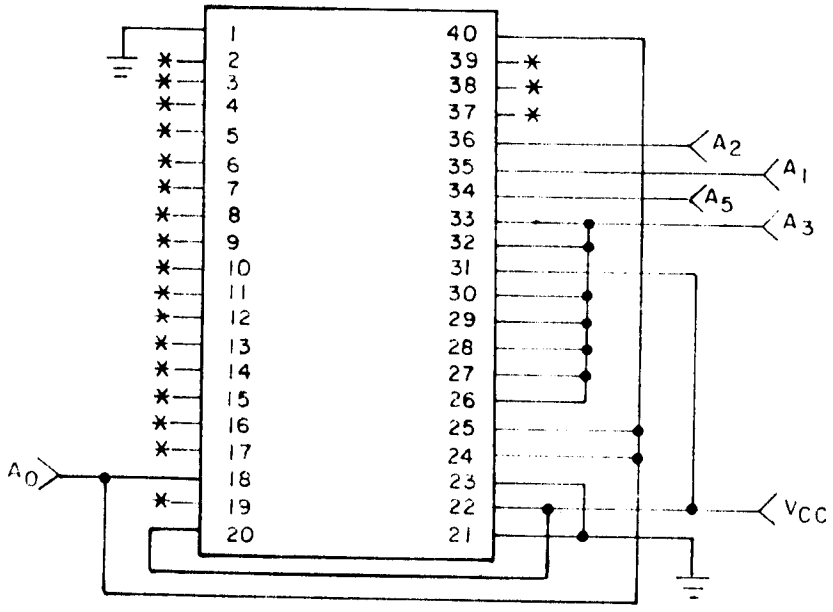


NOTE: $\Delta I_{supply} / \Delta T_A = -0.2\%/°C$.

FIGURE 5. Typical supply current versus temperature, normalized.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

Device type 01



V_{CC} = +5.0 Vdc
Ao = 100 kHz
Logic 1 = V_{CC}
Logic 0 = +0.3 Vdc
*Output open and isolated from other devices driven by same driver.

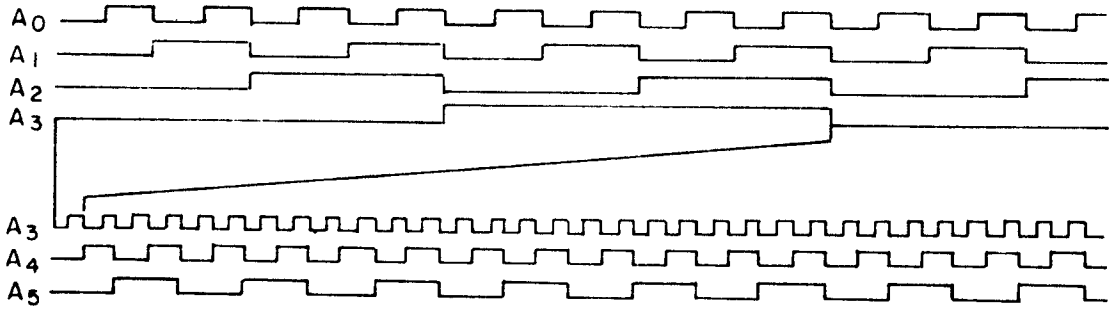
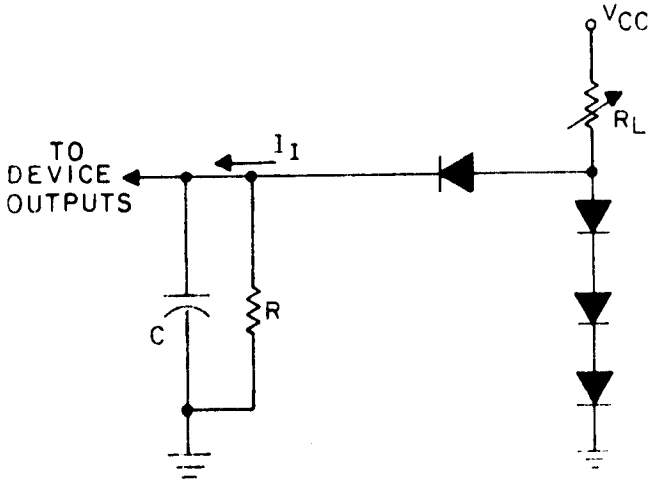


FIGURE 6. Burn-in and life test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.



NOTES:
1. C = 130 pF for pins 26-33.
2. C = 40 pF for pins 2-17, 19, 39.
3. R_L = 2.5 kΩ for pins 26-33.
4. R = 11.7 kΩ for pins 26-33.
5. R_L is adjusted so that I_I = 3.2 mA with V_I = 0.4 V for pins 2-17, 19, 39.
6. R = 12 kΩ for pins 2-17, 19, 39.

FIGURE 7. Dynamic functional test load circuit for DO-D7, PAO-PA7, PBO-PB7, CA2 and CB2.

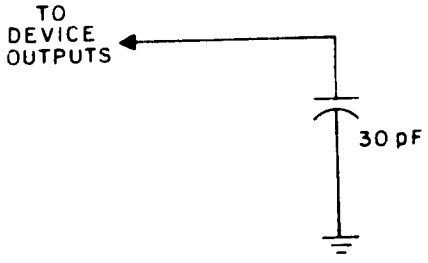


FIGURE 8. Dynamic functional test load circuit for PAO-PA7, CA2.

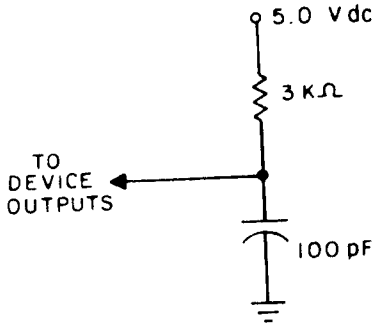


FIGURE 9. Dynamic functional test load circuit for IRQA, IRQB.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

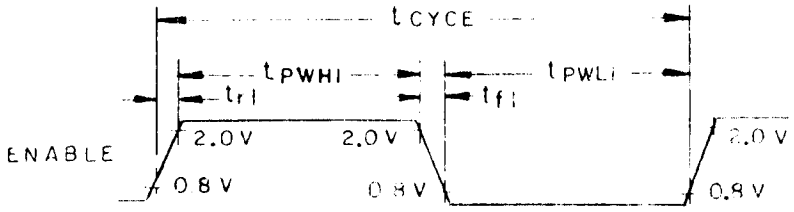


FIGURE 10. Enable signal characteristics.

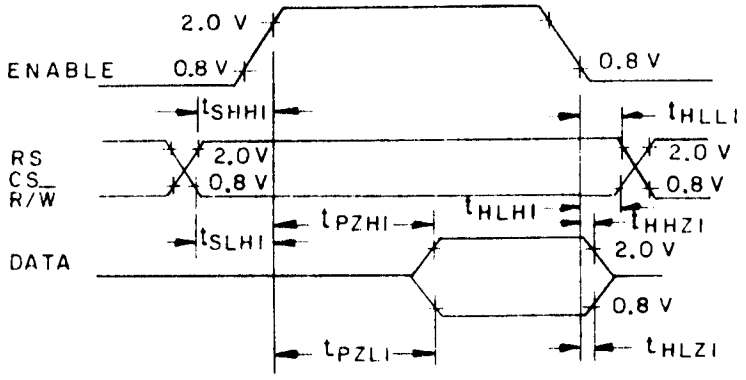


FIGURE 11. Bus read timing characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

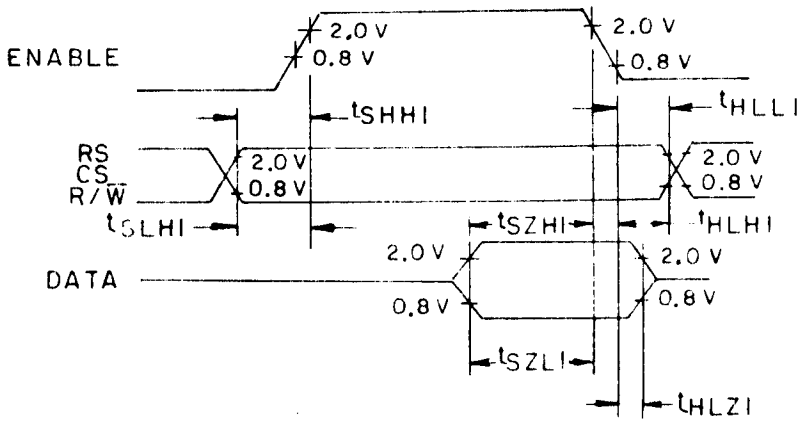


FIGURE 12. Bus write timing characteristics.

(Read Mode)

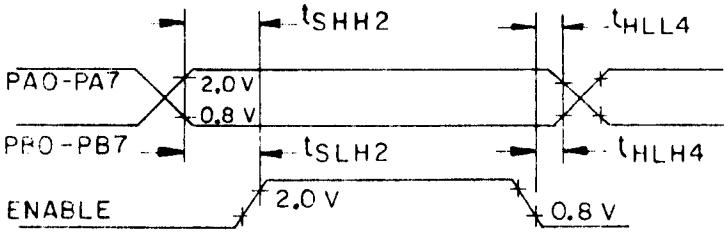


FIGURE 13. Peripheral data setup and hold times.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

(Read Mode: CRA5 = CRA3 = 1, CRA4 = 0)

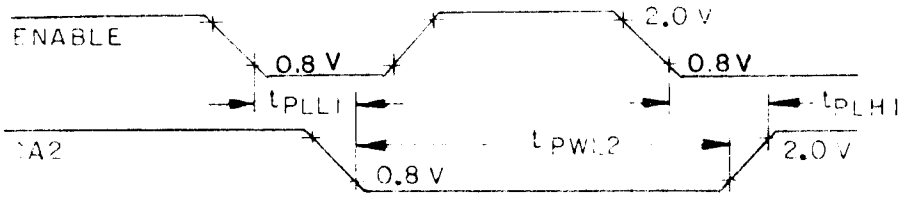


FIGURE 14. CA2 delay time.

(Read Mode: CRA5 = 1, CRA3 = CRA4 = 0)

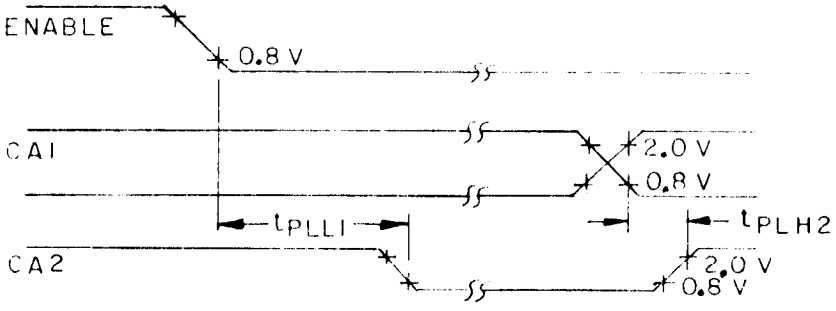


FIGURE 15. CA2 delay time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

(Write Mode: CRA5 = CRA3 = 1, CRA4 = 0)

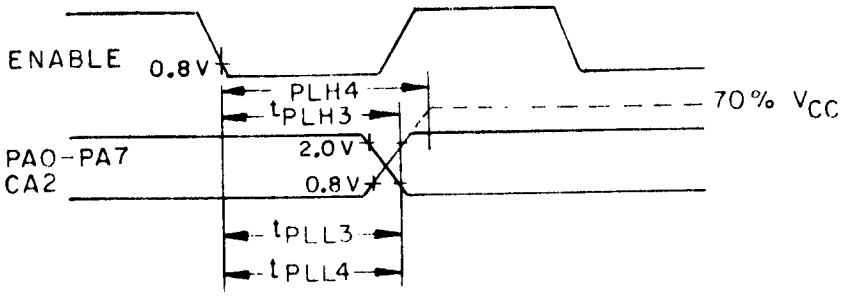


FIGURE 16. Peripheral data delay times.

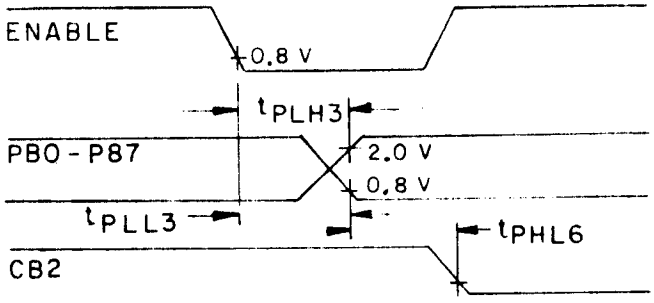


FIGURE 17. Peripheral data and CB2 delay times.

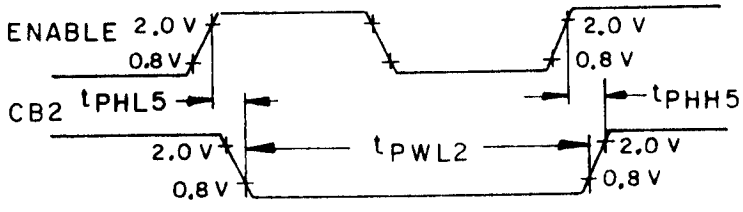


FIGURE 18. CB2 delay time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

(Write Mode: CRB5 = 1, CRB3 = CRB4 = 0)

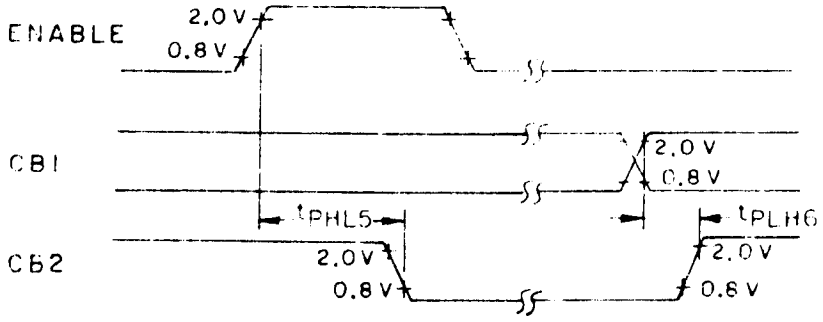


FIGURE 19. CB2 delay time.

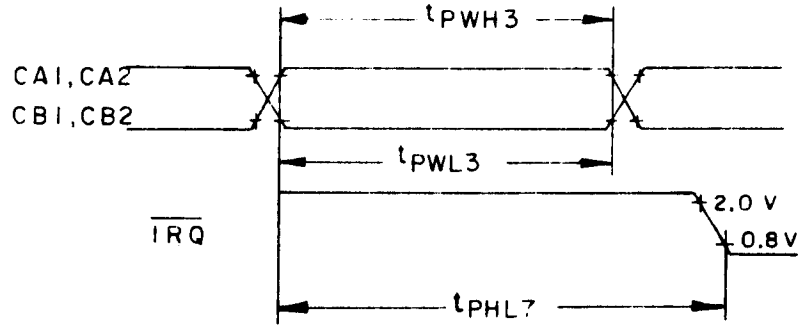


FIGURE 20. Interrupt pulse width and $\overline{IRQ}$ response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

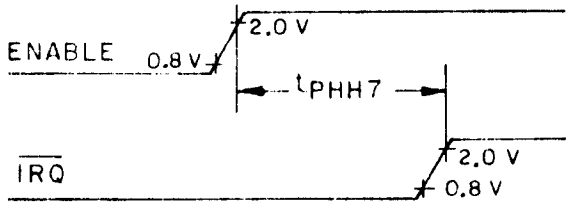
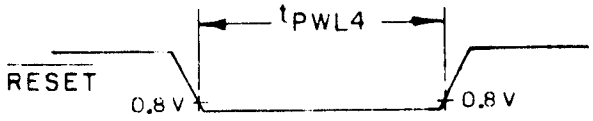


FIGURE 21. IRQ release time.



NOTE: The RESET line must be a V_{IL} for a minimum of 1.0 µs before addressing the PIA.

FIGURE 22. RESET low time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIA. Input conditions for table IIIC.

Symbol	Terminal	Pin number	Input conditions 1/			Figure
			C1	C2	C3	
V_{IH}	Logic inputs		2.0 V	2.0 V	2.0 V	
V_{IL}	Logic inputs		0.8 V	0.8 V	0.8 V	
t_{cycE}	ENABLE		1.0 µs	1.0 ms	2.0 µs	
t_{PWL1}	ENABLE	25	430 ns	430 ns	430 ns	7, 10
t_{PWL4}	RESET	34	1.0 µs	1.0 µs	1.0 µs	22
t_{SLH1}	$\overline{CS0}$, CS1, $\overline{CS2}$, R/$\overline{W}$, RS0, RS1	22, 23, 24, 21, 35, 36	160 ns	160 ns	160 ns	7, 11, 12
t_{SHH1}	"		160 ns	160 ns	160 ns	7, 11, 12
t_{HLH1}	"		10 ns	10 ns	10 ns	7, 11, 12
t_{HLL1}	"		10 ns	10 ns	10 ns	7, 11, 12
t_{SZH1} ($\overline{WRITE}$)	Data (IN)	26, 27, 28, 29, 30, 31, 32, 33	195 ns	195 ns	195 ns	7, 12
t_{SZL1} ($\overline{WRITE}$)	"		195 ns	195 ns	195 ns	7, 12
t_{HLH3} ($\overline{WRITE}$)	"		10 ns	10 ns	10 ns	7, 12
t_{HLL3} ($\overline{WRITE}$)	"	"	10 ns	10 ns	10 ns	7, 12

1/ See notes at end of table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIB. Functional test output conditions for subgroups 7 and 8 for table IIIC.

Symbol	Measured terminal	Figure	Test limits Subgroups 7 and 8 T_C = 25°, -55°, and 125°C		Unit
			Min	Max	
V_{OH}	All outputs		2.4		V
V_{OL}	All outputs			0.4	V
t_{PZL1} t_{PZH1}	DO-D7 DO-D7	7, 11 7, 11		320 320	ns
t_{PLH1}	CA2	7, 14		1,000	
t_{PLL1}	CA2	7, 14, 15		1,000	
t_{PLH2}	CA2	7, 15		2,000	
t_{PLH3}	CA2	7, 16, 17		1,000	
t_{PLL3}	CA2	7, 16, 17		1,000	
t_{PLH4}	PA0 - PA7 PB0 - PB7	8, 16		2,000	
t_{PLL4}	PA0 - PA7 PB0 - PB7	8, 16		2,000	
t_{PHH5}	PA0 - PA7 PB0 - PB7	7, 18		1,000	
t_{PHL5}	PA0 - PA7 PB0 - PB7	7, 18, 19		1,000	
t_{PLH6}	CB2	7, 19		2,000	
t_{PHL6}	CB2	7, 17	20		
t_{PHH7}	CB2	7, 21		1,600	
t_{PHL7}	CB2	7, 20		1,000	

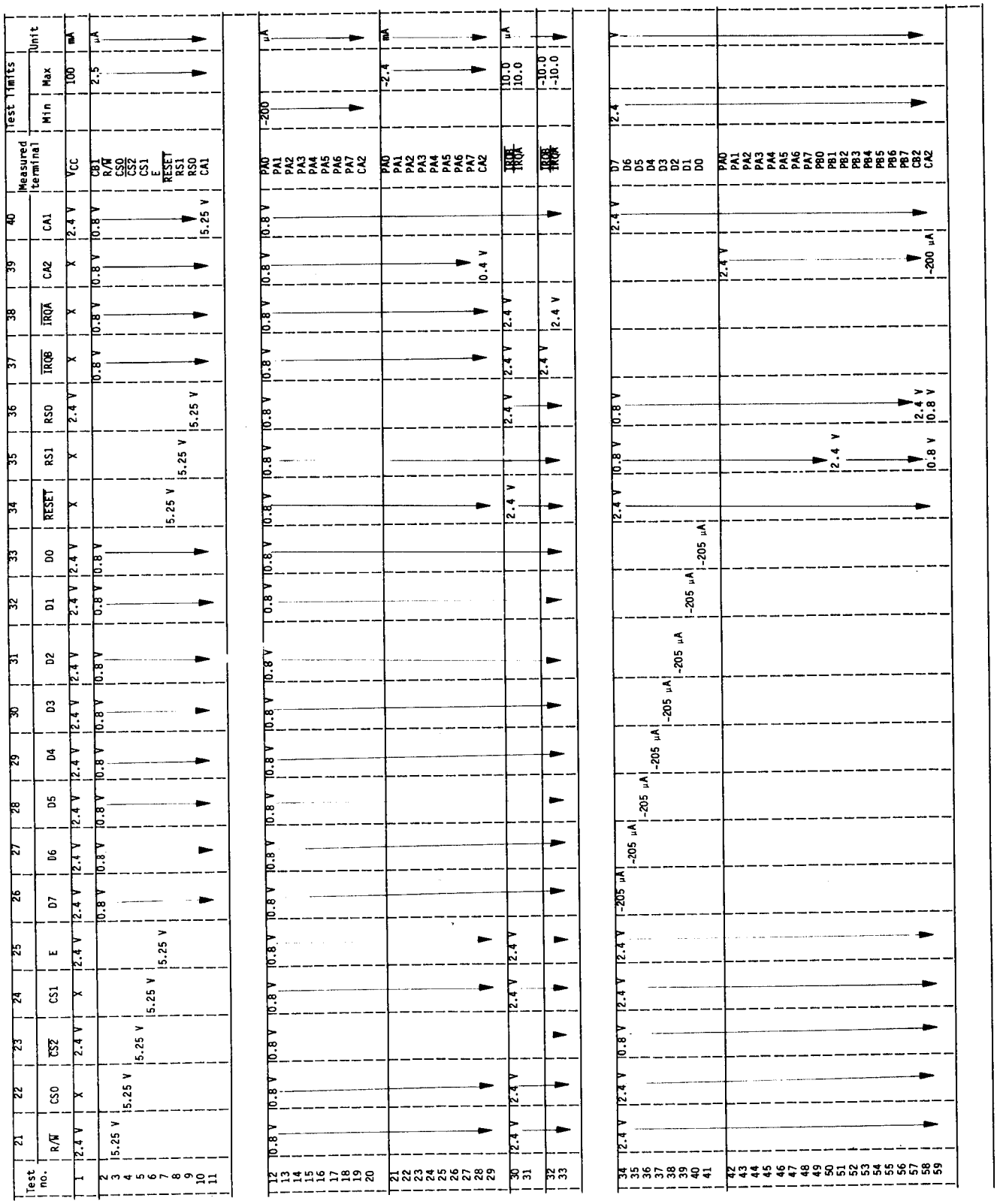
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIC. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				V_{SS}	PA0	PA1	PA2	PA3	PA4	PA5	PA6	PA7	PB0	PB1	PB2	PB3	PB4	PB5	PB6	PB7	CB1	CB2	V_{CC}
1 T_C = 25°C	I_{CC}	3005	1	GND	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	2.4 V	X	5.25 V
	I_{IH1}		2		0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	5.25 V 0.8 V	0.8 V	
			3																				
			4																				
			5																				
			6																				
			7																				
			8																				
			9																				
			10																				
			11																				
Run pattern 4 vectors 1-3, at vector 3 perform the following tests:																							
1 T_C = 25°C	I_{IH2}	3010	12	GND	2.4 V	2.4 V							0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	5.25 V
			13				2.4 V																
			14					2.4 V															
			15						2.4 V														
			16							2.4 V													
			17								2.4 V												
			18									2.4 V											
			19																				
			20																				
	I_{IL1}	3009	21		0.4 V																		
			22			0.4 V																	
			23				0.4 V																
			24					0.4 V															
			25						2.4 V 0.8 V														
			26		0.8 V	0.8 V	0.8 V	0.8 V		0.4 V													
			27								0.4 V												
			28									0.4 V											
			29																				
	I_{LOH} I_{LOH}		30							0.8 V	0.8 V	0.8 V										2.4 V	4.75 V
			31																				
	I_{LOL} I_{LOL}		32																				
			33																				
Run pattern 6 vectors 1-14, at vector 14 perform the following tests:																							
1 T_C = 25°C	V_{OH1}	3006	34	GND	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	2.4 V	4.75 V
			35																				
			36																				
			37																				
			38																				
			39																				
			40																				
			41																				
	V_{OH2}		42		-200 µA																		
			43			-200 µA																	
			44				-200 µA																
			45					-200 µA															
			46						-200 µA														
			47							-200 µA													
			48								-200 µA												
			49									-200 µA											
			50										-200 µA										
			51		0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V		-200 µA									
			52												-200 µA								
			53													-200 µA							
			54														-200 µA						
			55															-200 µA					
			56																-200 µA				
			57																	-200 µA			
			58										0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V			
			59	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V		-200 µA 2.4 V	
Run pattern 5 vectors 1-4, at vector 4 perform the following tests:																							

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIC. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)



Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits Min	Test limits Max	Unit
	R/$\overline{W}$	$\overline{CS0}$	$\overline{CS2}$	CS1	E	D7	D6	D5	D4	D3	D2	D1	D0	$\overline{RESET}$	RS1	RS0	$\overline{IRQB}$	$\overline{IRQA}$	CA2	CA1	V_{CC}		100	mA
1	2.4 V	X	2.4 V	X	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	X	X	2.4 V	X	X	X	2.4 V				
2	5.25 V					0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V				0.8 V	0.8 V	0.8 V	0.8 V	CB1		2.5	µA
3																					R/$\overline{W}$			
4		5.25 V																			$\overline{CS0}$			
5			5.25 V																		$\overline{CS2}$			
6				5.25 V																	CS1			
7					5.25 V									5.25 V							E			
8															5.25 V						RESET			
9																5.25 V					RS1			
10																					RS0			
11																				5.25 V	CA1			
12	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	PA0	-200		µA
13																					PA1			
14																					PA2			
15																					PA3			
16																					PA4			
17																					PA5			
18																					PA6			
19																					PA7			
20																					CA2			
21	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	PA0		-2.4	mA
22																					PA1			
23																					PA2			
24																					PA3			
25																					PA4			
26																					PA5			
27																					PA6			
28																					PA7			
29																					CA2			
30	2.4 V	2.4 V		2.4 V	2.4 V									2.4 V		2.4 V	2.4 V	2.4 V	2.4 V		$\overline{IRQB}$	10.0		µA
31																					$\overline{IRQA}$	10.0		
32																					$\overline{IRQB}$		-10.0	µA
33																					$\overline{IRQA}$		-10.0	
34	2.4 V	2.4 V	0.8 V	2.4 V	2.4 V	-205 µA								2.4 V	0.8 V	0.8 V				2.4 V	D7	2.4		V
35							-205 µA														D6			
36								-205 µA													D5			
37									-205 µA												D4			
38										-205 µA											D3			
39											-205 µA										D2			
40												-205 µA									D1			
41													-205 µA								D0			
42															2.4 V				2.4 V		PA0			
43																					PA1			
44																					PA2			
45																					PA3			
46																					PA4			
47																					PA5			
48																					PA6			
49																					PA7			
50																					PB0			
51																					PB1			
52																					PB2			
53																					PB3			
54																					PB4			
55																					PB5			
56																					PB6			
57																					PB7			
58																					CB2			
59																					CA2			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIC. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				V_{SS}	PA0	PA1	PA2	PA3	PA4	PA5	PA6	PA7	PB0	PB1	PB2	PB3	PB4	PB5	PB6	PB7	CB1	CB2	V_{CC}
1 T_C = 25°C	V_{OL1}	3007	60 61 62 63 64 65 66 67	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	4.75 V
Run pattern 6 vecotrs 1-29, at vector 29 perform the following tests:																							
1 T_C = 25°C	V_{OL2}	3007	68 69 70 71 72 73 74 75	GND	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	2.4 V	4.75 V
Run pattern 6 vectors 1-42, at vector 42 perform the following tests:																							
1 T_C = 25°C	V_{OL2}	3007	76 77 78 79 80 81 82 83 84	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	3.2 mA	2.4 V 3.2 mA	2.4 V	4.75 V
Run pattern 6 vectors 1-100, at vector 100 perform the following test:																							
1 T_C = 25°C	V_{OL2}	3007	85	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0. V	0.8 V	0.8 V	2.4 V	4.75 V
Run pattern 6 vectors 1-55, at vector 55 perform the following test:																							
1 T_C = 25°C	V_{OL2}	3007	86	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	2.4 V	4.75 V
Run pattern 6 vectors 1-215, at vector 215 perform the following tests:																							
1 T_C = 25°C	V_{OL2}	3007	87	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V		4.75 V
Run pattern 6 vectors 1-48, at vector 48 perform the following tests:																							
1 T_C = 25°C	I_{OH1}		88 89 90 91 92 93 94 95	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V	1.5 V	2.4 V	2.4 V	4.75 V

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIC. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)

Test no.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Measured terminal	Test limits		Unit
	R/$\overline{W}$	CS0	$\overline{CS2}$	CS1	E	D7	D6	D5	D4	D3	D2	D1	D0	$\overline{RESET}$	RS1	RSO	$\overline{IRQB}$	$\overline{IRQA}$	CA2	CA1		Min	Max	
60	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	1.6 mA	1.6 mA							0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	D7		0.4	V
61								1.6 mA													D6			
62									1.6 mA												D5			
63										1.6 mA											D4			
64											1.6 mA										D3			
65												1.6 mA									D2			
66													1.6 mA								D1			
67																					D0			
68	2.4 V	2.4 V	0.8 V	2.4 V	2.4 V									2.4 V	0.8 V	0.8 V			2.4 V	2.4 V	PA0		0.4	V
69																					PA1			
70																					PA2			
71																					PA3			
72																					PA4			
73																					PA5			
74																					PA6			
75																					PA7			
76	2.4 V	2.4 V	0.8 V	2.4 V	2.4 V									2.4 V	2.4 V	0.8 V			2.4 V	2.4 V	PB0		0.4	V
77																					PB1			
78																					PB2			
79																					PB3			
80																					PB4			
81																					PB5			
82																					PB6			
83																					PB7			
84																					CB1			
85	0.8 V	2.4 V	2.4 V	2.4 V	2.4 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	2.4 V	2.4 V	2.4 V			2.4 V	2.4 V	$\overline{IRQB}$		0.4	V
86	2.4 V	0.8 V	0.8 V	2.4 V	2.4 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	2.4 V	2.4 V	0.8 V			2.4 V	0.8 V	$\overline{IRQA}$		0.4	V
87	2.4 V	0.8 V	0.8 V	2.4 V	0.8 V	0.8 V	0.8 V	2.4 V	2.4 V	0.8 V	2.4 V	0.8 V	2.4 V	2.4 V	0.8 V	0.8 V				2.4 V	CA2		0.4	V
88	2.4 V	2.4 V	0.8 V	2.4 V	2.4 V	2.4 V								2.4 V	2.4 V	0.8 V			2.4 V	2.4 V	PB0		10.0	mA
89																					PB1			
90																					PB2			
91																					PB3			
92																					PB4			
93																					PB5			
94																					PB6			
95																					PB7			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIC . Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				V_{SS}	PA0	PA1	PA2	PA3	PA4	PA5	PA6	PA7	PB0	PB1	PB2	PB3	PB4	PB5	PB6	PB7	CB1	CB2	V_{CC}
Run pattern 5 vectors 1-20, at vector 20 perform the following test:																							
1 T_C = 25°C	I_{OH1}		96	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V			4.75 V
Run pattern 6 vectors 1-53, at vector 53 perform the following tests:																							
1 T_C = 25°C	I_{IL2}		97 98 99 100 101 102 103 104	GND	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	2.4 V	2.4 V	4.75 V
Run pattern 6 vectors 1-14, at vector 14 perform the following tests:																							
1 T_C = 25°C	I_{IL2}		105 106 107 108 109 110 111 112 113	GND	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	0.4 V	4.75 V
Run pattern 6 vectors 1-53, at vector 53 perform the following tests:																							
1 T_C = 25°C	I_{IH3}		114 115 116 117 118 119 120 121	GND	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V			4.75 V
Run pattern 6 vectors 1-14, at vector 14 perform the following tests:																							
1 T_C = 25°C	I_{IH3}		122 123 124 125 126 127 128 129 130	GND	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	2.4 V	2.4 V	4.75 V
	V_{OH3}		131 132 133 134 135 136 137 138 139	GND	-10 µA	-10 µA	-10 µA	-10 µA	-10 µA	-10 µA	-10 µA	-10 µA												

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIC. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)

Test no.	21 R/$\overline{W}$	22 CS0	23 $\overline{CS2}$	24 CS1	25 E	26 D7	27 D6	28 D5	29 D4	30 D3	31 D2	32 D1	33 DO	34 $\overline{RESET}$	35 RS1	36 RS0	37 $\overline{IRQB}$	38 $\overline{IRQA}$	39 CA2	40 CA1	Measured terminal	Test limits Min	Test limits Max	Unit
96	2.4 V	2.4 V	0.8 V	2.4 V	2.4 V									2.4 V	2.4 V	2.4 V			2.4 V	2.4 V	CB2		10.0	mA
97	2.4 V	2.4 V	0.8 V	0.8 V	2.4 V									2.4 V	2.4 V	0.8 V			2.4 V	0.8 V	PB0		10.0	µA
98	↓	↓	↓	↓	↓									↓	↓	↓			↓	↓	PB1		↓	↓
99																					PB2			
100																					PB3			
101																					PB4			
102																					PB5			
103																					PB6			
104																					PB7			
105	2.4 V	2.4 V	2.4 V	0.8 V	2.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	2.4 V	0.8 V	0.8 V				2.4 V	CB2		10.0	µA
106	↓	↓	↓	↓	↓									↓	↓	↓				↓	D7		↓	↓
107																					D6			
108																					D5			
109																					D4			
110																					D3			
111																					D2			
112																					D1			
113																					DO			
114																					PB0		10.0	µA
115																					PB1		↓	↓
116																					PB2			
117																					PB3			
118																					PB4			
119																					PB5			
120																					PB6			
121																					PB7			
122	2.4 V	2.4 V	2.4 V	0.8 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	0.8 V	0.8 v				2.4 V	CB2		10.0	µA
123	↓	↓	↓	↓	↓									↓	↓	↓				↓	D7		↓	↓
124																					D6			
125																					D5			
126																					D4			
127																					D3			
128																					D2			
129																					D1			
130																					DO			
131																			-10 µA		PA0		3.75	V
132																			↓		PA1		↓	↓
133																					PA2			
134																					PA3			
135																					PA4			
136																					PA5			
137																					PA6			
138																					PA7			
139																					CA2			

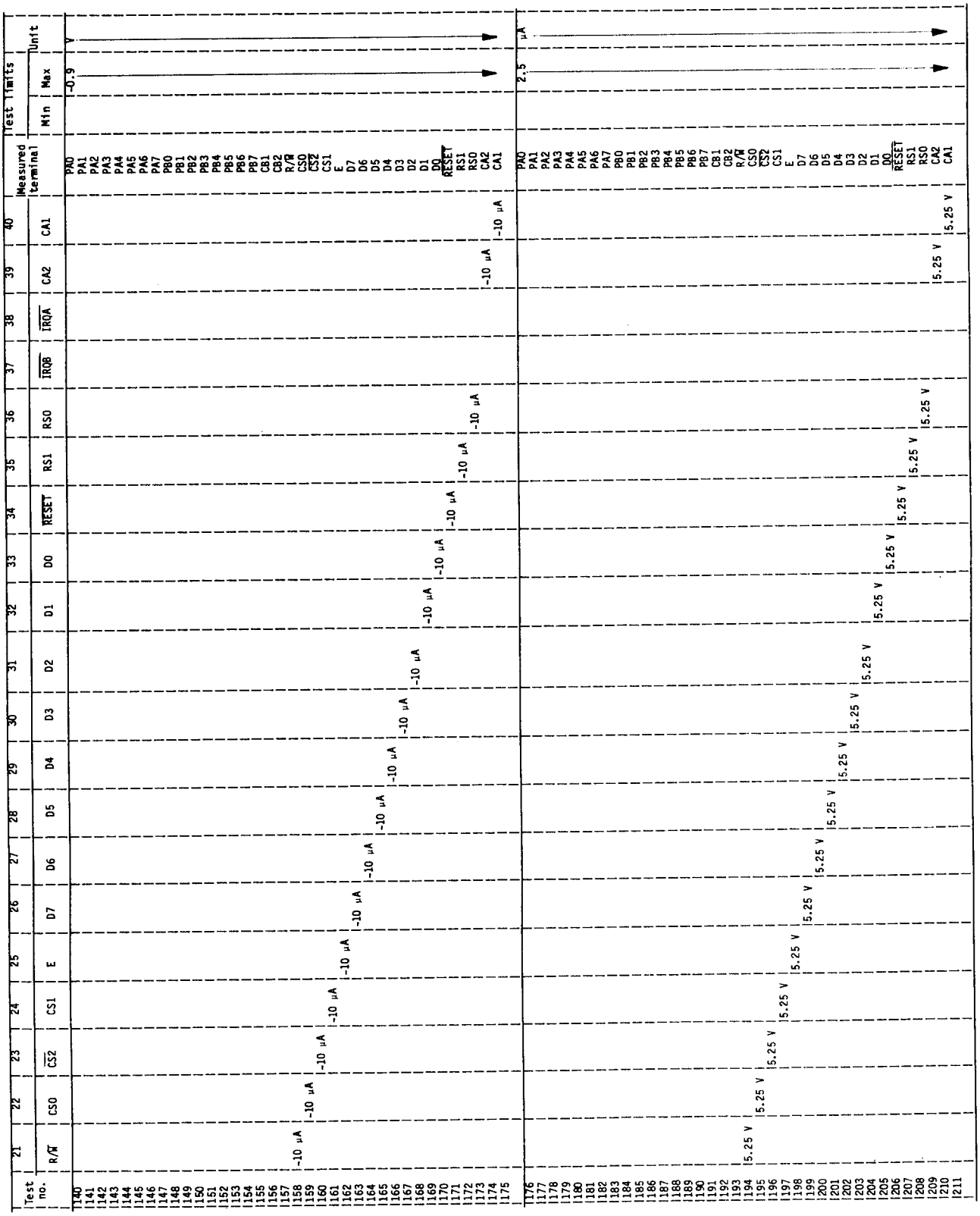
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				V_{SS}	PA0	PA1	PA2	PA3	PA4	PA5	PA6	PA7	PB0	PB1	PB2	PB3	PB4	PB5	PB6	PB7	CB1	CB2	V_{CC}
1	V_{IC} (NEG)		140	GND	-10 µA																		GND
T_C = 25°C			141			-10 µA																	
			142				-10 µA																
			143					-10 µA															
			144						-10 µA														
			145							-10 µA													
			146								-10 µA												
			147									-10 µA											
			148										-10 µA										
			149											-10 µA									
			150												-10 µA								
			151													-10 µA							
			152														-10 µA						
			153															-10 µA					
			154																-10 µA				
			155																	-10 µA			
			156																		-10 µA		
			157																			-10 µA	
			158																				-10 µA
			159																				
			160																				
			161																				
			162																				
			163																				
			164																				
			165																				
			166																				
			167																				
			168																				
			169																				
			170																				
			171																				
			172																				
			173																				
			174																				
			175																				
	V_{IC} (POS)		176		5.25 V																		
			177			5.25 V																	
			178				5.25 V																
			179					5.25 V															
			180						5.25 V														
			181							5.25 V													
			182								5.25 V												
			183									5.25 V											
			184										5.25 V										
			185											5.25 V									
			186												5.25 V								
			187													5.25 V							
			188														5.25 V						
			189															5.25 V					
			190																5.25 V				
			191																	5.25 V			
			192																		5.25 V		
			193																			5.25 V	
			194																				5.25 V
			195																				
			196																				
			197																				
			198																				
			199																				
			200																				
			201																				
			202																				
			203																				
			204																				
			205																				
			206																				
			207																				
			208																				
			209																				
			210																				
			211																				

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)



Test no.	21 R/W	22 CS0	23 CS2	24 CS1	25 E	26 D7	27 D6	28 D5	29 D4	30 D3	31 D2	32 D1	33 D0	34 RESET	35 RS1	36 RS0	37 IRQB	38 IRQA	39 CA2	40 CA1	Measured terminal	Test limits Min	Test limits Max	Unit
140																					PA0		-0.9	V
141																					PA1			
142																					PA2			
143																					PA3			
144																					PA4			
145																					PA5			
146																					PA6			
147																					PA7			
148																					PB0			
149																					PB1			
150																					PB2			
151																					PB3			
152																					PB4			
153																					PB5			
154																					PB6			
155																					PB7			
156																					CB1			
157																					CB2			
158	-10 µA																				R/W			
159		-10 µA																			CS0			
160			-10 µA																		CS2			
161				-10 µA																	CS1			
162					-10 µA																E			
163						-10 µA															D7			
164							-10 µA														D6			
165								-10 µA													D5			
166									-10 µA												D4			
167										-10 µA											D3			
168											-10 µA										D2			
169												-10 µA									D1			
170													-10 µA								D0			
171														-10 µA							RESET			
172															-10 µA						RS1			
173																-10 µA					RS0			
174																			-10 µA		CA2			
175																				-10 µA	CA1			
176																					PA0		2.5	µA
177																					PA1			
178																					PA2			
179																					PA3			
180																					PA4			
181																					PA5			
182																					PA6			
183																					PA7			
184																					PB0			
185																					PB1			
186																					PB2			
187																					PB3			
188																					PB4			
189																					PB5			
190																					PB6			
191																					PB7			
192																					CB1			
193	5.25 V																				CB2			
194		5.25 V																			R/W			
195			5.25 V																		CS0			
196				5.25 V																	CS2			
197					5.25 V																CS1			
198						5.25 V															E			
199							5.25 V														D7			
200								5.25 V													D6			
201									5.25 V												D5			
202										5.25 V											D4			
203											5.25 V										D3			
204												5.25 V									D2			
205													5.25 V								D1			
206														5.25 V							D0			
207															5.25 V						RESET			
208																5.25 V					RS1			
209																			5.25 V		RS0			
210																				5.25 V	CA2			
211																					CA1			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

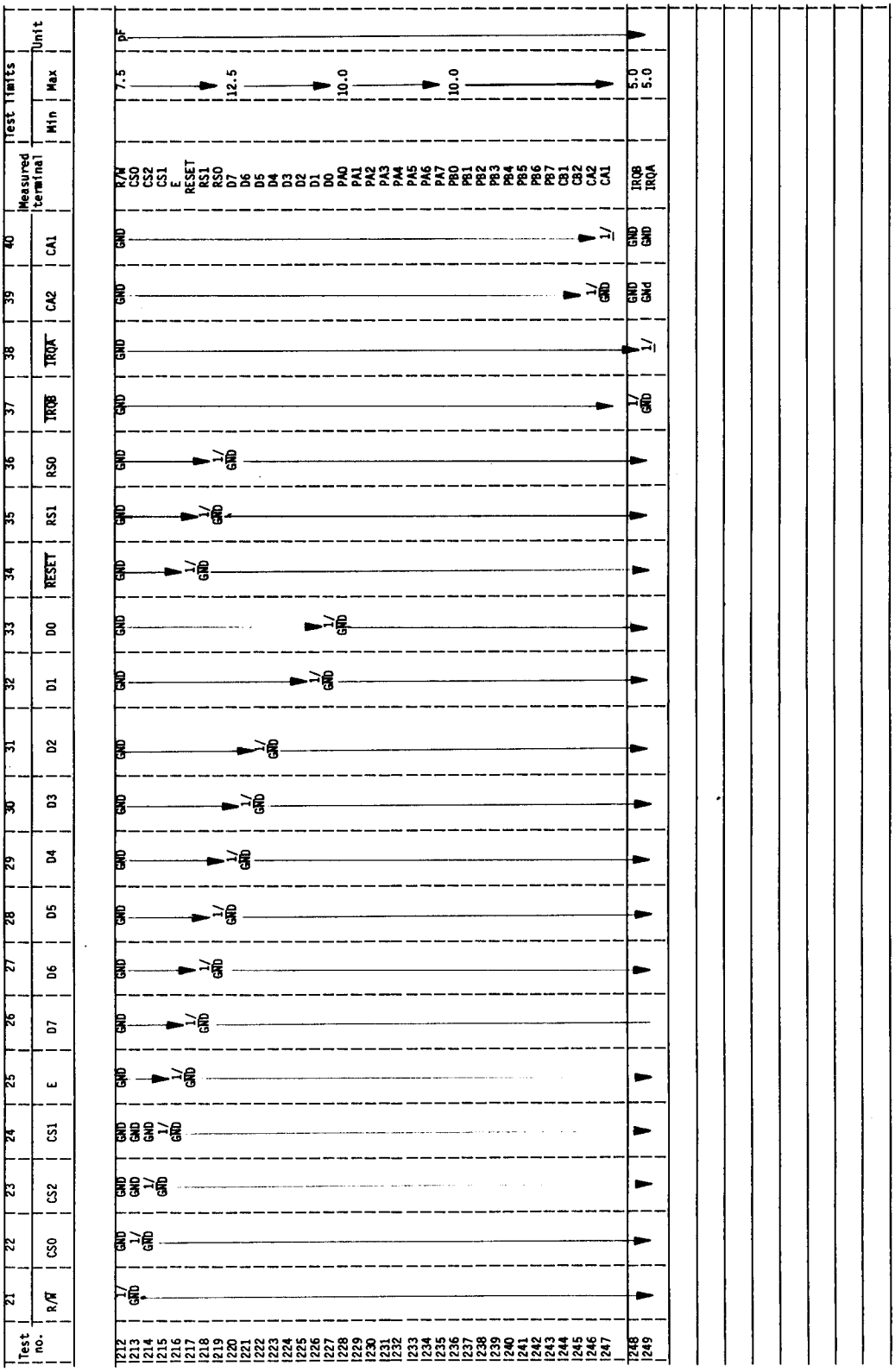
TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V or open)

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				V_{SS}	PA0	PA1	PA2	PA3	PA4	PA5	PA6	PA7	PB0	PB1	PB2	PB3	PB4	PB5	PB6	PB7	CB1	CB2	V_{CC}
2	Repeat subgroup 1 tests 1-139 at T_C = +125°C.																						
4 T_C = 25°C	I_{CC1}	3012	212	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND
			213																				
			214																				
			215																				
			216																				
			217																				
			218																				
			219																				
			220																				
			221																				
			222																				
			223																				
			224																				
			225																				
			226																				
			227		1/ GND																		
			228			1/ GND																	
			229				1/ GND																
			230					1/ GND															
			231						1/ GND														
			232							1/ GND													
			233								1/ GND												
			234									1/ GND											
			235										1/ GND										
			236											1/ GND									
			237												1/ GND								
			238													1/ GND							
			239														1/ GND						
			240															1/ GND					
			241																1/ GND				
			242																	1/ GND			
			243																		1/ GND		
			244																			1/ GND	
			245																				
			246																				
			247																				
	I_{OUT1}		248																				
	I_{OUT2}		249																				
7 T_C = 25°C	Dynamic, functional tests	3014, 3007, 3006, 3003		Table IV, pattern 1 - Input test condition C1 of table IIIA with V_{CC} = 4.75 Vdc.																			
				Table IV, pattern 1 - Input test condition C1 of table IIIA with V_{CC} = 5.25 Vdc.																			
				Table IV, pattern 1 - Input test condition C2 of table IIIA with V_{CC} = 4.75 Vdc.																			
				Table IV, pattern 1 - Input test condition C2 of table IIIA with V_{CC} = 5.25 Vdc																			
				Table IV, pattern 2 and 3 - Input test condition C3 of table IIIA with V_{CC} = 4.75 Vdc.																			
				Table IV, pattern 2 and 3 - Input test condition C3 of table IIIA with V_{CC} = 5.25 Vdc.																			
8	Repeat subgroup 7, at -55°C.																						
8	Repeat subgroup 7 at +125°C.																						

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V; or low ≤0.8 V or open)



Test no.	21 R/W	22 CS0	23 CS2	24 CS1	25 E	26 D7	27 D6	28 D5	29 D4	30 D3	31 D2	32 D1	33 D0	34 RESET	35 RS1	36 RS0	37 IRQB	38 IRQA	39 CA2	40 CA1	Measured terminal	Test limits Min	Test limits Max	Unit
212	1/ GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	R/W		7.5	pF
213		1/ GND	GND	GND																	CS0			
214			1/ GND	GND																	CS2			
215				1/ GND	1/ GND																CS1			
216																					E			
217						1/ GND								1/ GND							RESET		12.5	
218							1/ GND								1/ GND						RS1			
219								1/ GND								1/ GND					RS0			
220									1/ GND												D7			
221										1/ GND											D6		10.0	
222											1/ GND										D5			
223												1/ GND									D4			
224													1/ GND								D3			
225																					D2			
226																					D1			
227																					D0			
228																					PA0		10.0	
229																					PA1			
230																					PA2			
231																					PA3			
232																					PA4			
233																					PA5			
234																					PA6			
235																					PA7		10.0	
236																					PB0			
237																					PB1			
238																					PB2			
239																					PB3			
240																					PB4			
241																					PB5			
242																					PB6			
243																					PB7			
244																					CB1			
245																					CB2			
246																					CA2			
247																					CA1			
248																					IRQB		5.0	
249																					IRQA		5.0	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IIIC. Group A inspection - Continued.

Symbol	MIL-STD-883 method	Measurement Location: Pattern number	Input condition from table IIIA	Waveform figure	Parameter verified	Measured terminal: Input under test	Measured terminal: Output under test	Test Limits 2/ Subgroup 12 T_C = +125°C: Min	Test Limits 2/ Subgroup 12 T_C = +125°C: Max	Unit
t_{PZH1} t_{PZL1}	3003, 3014	1	C1	7, 11		ENABLE	D0 - D7		320	ns
t_{PLL1}				7, 14, 15		ENABLE	CA2		1,000	
t_{PLH1}				7, 14		ENABLE	CA2		1,000	
t_{PLH2}				7, 15		CA1	CA2		2,000	
t_{PLH3} t_{PLL3}				7, 16, 17		ENABLE	PA0 - PA7 CA2		1,000	
t_{PLH4} t_{PLL4}				8, 16		ENABLE	PA0 - PA7 PB0 - PB7		2,000	
t_{PHH5} t_{PHL5}				7, 18, 19		ENABLE	CB2		1,000	
t_{PLH6}				7, 19		PB0 - PB7	CB2		2,000	
t_{PHL6}				7, 17		PB0 - PB7	CB2	20		
t_{PHH7}				9, 21		ENABLE	CB2		1600	
t_{PHL7}				9, 20		CB1	CB2		1,000	
t_{PLH4} t_{PLL4}		2	C2	8, 16		ENABLE	PA0 - PA7 CA2		2,000	
V_{OH}	3006, 3014		C1				PA0 - PB7 PB0 - PB7	2.4		
							CB2			
							D0 - D7			
							$\overline{IRQB}$			
							IRQA			
							CA2			
V_{OL}	3007, 3014						PA0 - PA7 PB0 - PB7		0.4	
							CB2			
							D0 - D7			
							$\overline{IRQB}$			
							IRQA			
							CA2			

1/ The capacitance measurement shall be made between the indicated terminal and the grounded terminals at a frequency of 1 MHz. The dc bias of the measuring instrument shall be less than +0.1 V. The ac signal amplitude shall be less than 50 mVrms.

2/ Where propagation delay tests for subgroup 12 indicate a group of pins and where two or more propagation delays are specified for a test, only the reading for the pin with the worst delay requires recording. In some cases this will be a minimum reading, while for others it will be a maximum reading as indicated in the table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

TABLE IV. Vectors.

TABLE IV, not printed herein, is a computer generated listing for use with table III. Each vector identifies activity states by pin number. See section 6 for information on obtaining the complete table in compatible output format.

NOTES:

1/ Table IV, patterns 1, 2, and 3, when performed as specified in table IIIC for subgroups 7 and 8, verifies the dynamic, functional output level, threshold, and switching parameters at 25°C, 125°C, and -55°C.

2/ Input States shall be:

0 = Driven logic level low with comparator unmasked (for bidirectional pins).
1 = Driven logic level high with comparator unmasked (for bidirectional pins).
2 = Driven input logic level low with comparator masked.
3 = Driven input logic level high with comparator masked.

3/ Output States shall be:

X = A masked output compare with logic level low.
Y = A masked output compare with logic level high.
L = An unmasked output compare with logic level low.
H = An unmasked output compare with logic level high.
Z = High impedance; 1.0 V minimum to 1.5 V maximum with the load shown in figures 7-9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Computer generated table. Table IV, which forms a part of this specification in not printed herein because of its extreme length; and even when it is presented as a hard copy it requires conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be obtained in a form compatible with test system architecture. The preparing activity (or Defense Electronics Supply Center, as its agent) may be consulted for information on obtaining such output.

6.3 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.4 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase or direct shipment to the Government.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

Peripheral interface adapter bidirectional data (D0-D7) - The bidirectional data lines (D0-D7) allow the transfer of data between the microprocessor and the peripheral interface adapter.

Peripheral interface adapter enable (E) - The enable pulse, E, is the only timing signal that is supplied to the peripheral interface adapter. Timing of all other signals is referenced to the leading and trailing edges of the E pulse.

Peripheral interface adapter read/write (R/W) - This signal is generated by the microprocessor to control the direction of data transfers on the data bus.

$\overline{\text{Reset}}$ - The active low reset line is used to reset all register bits in the peripheral interface adapter to a logical zero (low). This line can be used as a power-on reset and as a master reset during system operation.

Peripheral interface adapter chip select (CS0, CS1, and CS2) These three input signals are used to select the peripheral interface adapter.

Peripheral interface adapter register select (RS0 and RS1) - The two register select lines are used to select the various registers inside the peripheral interface adapter. These two lines are used in conjunction with internal control registers to select a particular register that is to be written or read.

Interrupt request ($\overline{\text{IRQA}}$ and $\overline{\text{IRQB}}$) - The active low interrupt request lines, act to interrupt the microprocessor either directly or through interrupt priority circuitry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

Each interrupt request line has two internal interrupt flag bits that can cause the interrupt request line to go low. Each flag bit is associated with a particular peripheral interrupt line. Also four interrupt enable bits are provided in the peripheral interface adapter which may be used to inhibit a particular interrupt from a peripheral device.

Section A peripheral data (PA0-PA7) - Each of the peripheral data lines can be programmed to act as an input or output. This is accomplished by setting a "1" in the corresponding data direction register bit for those lines which are to be outputs. An "0" in a bit of the data direction register causes the corresponding peripheral data line to act as an input. During an MPU read peripheral data operation, the data on peripheral lines programmed to act as inputs appears directly on the corresponding MPU data bus lines.

Section B peripheral data (PB0-PB7) - The peripheral data lines in the B section of the PIA can be programmed to act as either inputs or outputs in a similar manner to PA0-PA7. However, the output buffers driving these lines differ from those driving lines PA0-PA7. They have three-state capability, allowing them to enter a high impedance state when the peripheral data line is used as an input.

Interrupt input (CA1 and CB1) - Peripheral input lines CA1 and CB1 are input only lines, that set the interrupt flags of the control registers. The active transition for these signals is also programmed by the two control registers.

Peripheral control (CA2 and CB2) - Peripheral control lines CA2 and CB2 can be programmed to act as interrupt inputs or as peripheral control outputs.

6.6 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finsih C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.7 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.4) generic data shall be on date codes no more than one year old and on the same package type see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification shall functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	MC6821

6.9 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protection devices have been designated in the chip to minimize the effect of this static buildup. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surfaces.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.10 Ordering guidance. Since the qualification and certification requirements have been removed from the specification, orders may be placed immediately.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 19, 85, 99
Navy - OS, SH
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0603-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

APPENDIX

INTERNAL CONTROLS,
PERIPHERAL INTERFACE ADAPTER

10. SCOPE

10.1 Scope. This appendix describes the internal controls of the peripheral interface adapter covered by this specification.

20. APPLICABLE DOCUMENTS.

20.1 This section is not applicable to this appendix.

30. INTERNAL CONTROLS. There are six locations within the PIA accessible to the MPU data bus: Two peripheral registers, two data direction registers, and two control registers. Selection of these locations is controlled by the RS0 and RS1 inputs together with bit 2 in the control register (see figure 23).

		Control register bit		
RS1	RS2	CRA-2	CRB-2	Location selected
0	0	1	X	Peripheral register A
0	0	0	X	Data direction register A
0	1	X	X	Control register A
1	0	X	1	Peripheral register B
1	0	X	0	Data direction register B
1	1	X	X	Control register B

X = Don't care

FIGURE 23. Internal addressing.

30.1 Initialization. A low reset line has the effect of zeroing all PIA registers. This will set PA0-PA7, PB0-PB7, CA2 and CB2 as inputs, and all interrupts disabled. The PIA must be configured during the restart program which follows the reset. Details of possible configurations of the data direction and control register are described in 20.2 and 20.3.

30.2 Data direction registers. The two data direction registers (DDRA and DDRB) allow the MPU to control the direction of data through each corresponding peripheral data line. A data direction register bit set at "0" configures the corresponding peripheral data line as an input; a "1" results in an output.

30.3 Control registers. The two control registers (CRA and CRB) allow the MPU to control the operation of the four peripheral control lines CA1, CA2, CB1, and CB2. In addition, they allow the MPU to enable the interrupt lines and monitor the status of the interrupt flags. Bits 0 through 5 of the two registers may be written or read by the MPU when the proper chip select and register select signals are applied. Bits 6 and 7 of the two registers are read only and are modified by external interrupts occurring on control lines CA1, CA2, CB1, and CB2 (see figure 24).

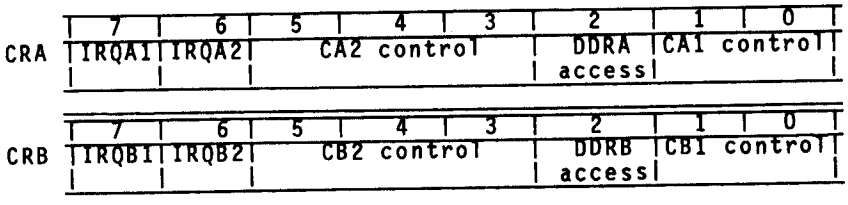


FIGURE 24. Control word format.

30.3.1 Data direction access control bit (CRA-2 and CRB-2). Bit 2 in each control register (CRA and CRB) allows selection of either a peripheral interface register or the data direction register when the proper register select signals are applied to RS0 and RS1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

30.3.2 Interrupt flags (CRA-6, CRA-7, CRB-6, and CRB-7). The four interrupt flag bits are set by active transitions of signals on the four interrupt and peripheral control lines when those lines are programmed to be inputs. These bits cannot be set directly from the MPU data bus and are reset indirectly by a read peripheral data operation on the appropriate section.

30.3.3 Control of CA1 and CB1 interrupt input lines (CRA-0, CRA-1, CRB-0, and CRB-1). The two lowest order bits of the control registers are used to control the interrupt input lines CA1 and CB1. Bits CRA-0 and CRB-0 are used to enable the MPU interrupt signals IRQA and IRQB, respectively. Bits CRA-1 and CRB-1 determine the active transition of the interrupt input signals CA1 and CB1 (see figures 25 and 26).

CRA-1 (CRB-1)	CRA-0 (CRB-0)	Interrupt input CA1 (CB1)	Interrupt flag CRA-7 (CRB-7)	MPU interrupt request $\overline{IRQA}$ ($\overline{IRQB}$)
0	0	Active	Set high on ↓ of CA1 (CB1)	Disabled $\overline{IRQ}$ remains high
0	1	Active	Set high on ↓ of CA1 (CB1)	Goes low when the interrupt flag bit CRA-7 (CRB-7) goes high
1	0	Active	Set high on ↑ of CA1 (CB1)	Disabled $\overline{IRQ}$ remains high
1	1	Active	Set high on ↑ of CA1 (CB1)	Goes low when the interrupt flag bit CRA-7 (CRB-7) goes high

NOTES:
1. ↑ indicates positive transition (low to high).
2. ↓ indicates negative transition (high to low).
3. The interrupt flag bit CRA-7 is cleared by an MPU read of the A data register and CRB-7 is cleared by an MPU read of the B data register.
4. If CRA-0 (CRB-0) is low when an interrupt occurs (interrupt disabled) and is later brought high, $\overline{IRQA}$ ($\overline{IRQB}$) occurs after CRA-0 (CRB-0) is written to a "one".

FIGURE 25. Control of interrupt inputs CA1 and CB1.

CRA-5 (CRB-5)	CRA-4 (CRB-4)	CRA-3 (CRB-3)	Interrupt input CA2 (CB2)	Interrupt flag CRA-6 (CRB-6)	MPU interrupt request $\overline{IRQA}$ ($\overline{IRQB}$)
0	0	0	Active	Set high on ↓ of CA2 (CB2)	Disabled $\overline{IRQ}$ remains high
0	0	1	Active	Set high on ↓ of CA2 (CB2)	Goes low when the interrupt flag bit CRA-6 (CRB-6) goes high
0	1	0	Active	Set high on ↑ of CA2 (CB2)	Disabled $\overline{IRQ}$ remains high
0	1	1	Active	Set high on ↑ of CA2 (CB2)	Goes low when the interrupt flag bit CRA-6 (CRB-6) goes high

NOTES:
1. ↑ indicates positive transition (low to high).
2. ↓ indicates negative transition (high to low).
3. The interrupt flag bit CRA-6 is cleared by an MPU read of the A data register and CRB-6 is cleared by an MPU read of the B data register.
4. If CRA-3 (CRB-3) is low when an interrupt occurs (interrupt disabled) and is later brought high, $\overline{IRQA}$ ($\overline{IRQB}$) occurs after CRA-3 (CRB-3) is written to a "one".

FIGURE 26. Control of CA2 and CB2 as an interrupt inputs. CRA5 (CRB5) is low

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.

30.3.4 Control of CA2 and CB2 Peripheral Control Lines (CRA-3, 4, 5 and CRB-3, 4, 5). Bits 3, 4, and 5 of the two control registers are used to control the CA2 and CB2 peripheral control lines. These bits determine if the control lines will be an interrupt input or an output control signal. If bit CRA-5 (CRB-5) is low, CA2 (CB2) is an interrupt input line similar to CA1 (CB1) (see figure 27). When CRA-5 (CRB-5) is high CA2 (CBA2) becomes an output signal that may be used to control peripheral data transfers. When in the output mode, CA2 and CB2 have slightly different characteristics (see figures 27 and 28).

			CB2	
CRB-5	CRB-4	CRB-3	Cleared	Set
1	0	0	Low on the positive transition of the first E pulse following an MPU write "B" data register operation	High when the interrupt flag bit CRB-7 is set by an active transition of the CB1 signal
1	0	1	Low on the positive transition of the first E pulse after an MPU write "B" data register operation	High on the positive edge of the first "E" pulse following an "E" pulse which occurred while the part was deselected
1	1	0	Low when CRB-3 goes low as a result of an MPU write in control register "B"	Always low as long as CRB-3 is low. Will go high on an MPU write in control register "B" that changes CRB-3 one.
1	1	1	Always high as long as CRB-3 is high. Will be cleared when an MPU write control register "B" results in clearing CRB-3 to zero	High when CRB-3 goes high as a result of an MPU write into control register "B"

FIGURE 27. Control of CB2 as an output.
CRB-5 is high

			CA2	
CRA-5	CRA-4	CRA-3	Cleared	Set
1	0	0	Low on the negative transition of E after an MPU read "A" data operation	High when the interrupt flag bit CRA-7 is set by an active transition of the CA1 signal
1	0	1	Low on the negative transition of E after an MPU read "A" data operation	High on the negative edge of the first "E" pulse which occurs during a deselect
1	1	0	Low when CRA-3 goes low as a result of an MPU write to control register "A"	Always low as long as CRA-3 is low. Will go high on an MPU write to control register "A" that changes CRA-3 to "one"
1	1	1	Always high as long as CRA-3 is high. Will be cleared on an MPU write control register "A" that clears CRA-3 to a "zero"	High when CRA-3 goes high as a result of an MPU write to control register "A"

FIGURE 28. Control of CA2 as an output.
CRA-5 is high

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:37Z
Check the source to verify that this is the current version before use.